

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 17 September 2026, 8:18 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:21 AM**Time taken** 2 mins 24 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main() {
4      int n, rev = 0, remainder;
5      int count = 0; // Counter variable to track operations
6
7      // Read input silently without any printf prompt strings
8      if (scanf("%d", &n) != 1) return 0;
9
10     // --- Counter Method Implementation ---
11
12     rev = 0;
13     count++; // For: rev = 0;
14
15     count++; // For the initial condition check of the while loop (n != 0)
16     while (n != 0) {
17         remainder = n % 10;
18         count++; // For: remainder = n % 10;
19
20         rev = rev * 10 + remainder;
21         count++; // For: rev = rev * 10 + remainder;
22
23         n /= 10;
24         count++; // For: n /= 10;
25
26         count++; // For the subsequent condition check of the while loop
27     }
28
29     count++; // For: print(rev); (Conceptual print operation)
30
31     // Output only the final value of the counter variable
32     printf("%d\n", count);
33
34     return 0;
35 }
36
```

37

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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